

Project Presentation of Industry Oriented Hands on Experience (22CS421)

On

Crop Yield Prediction Using Machine Learning

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Introduction: Why Crop Yield Prediction Matters

The Challenge

- **Food security:** India feeds 1.4 billion people; unpredictable yields threaten supply chains
- **Climate variability:** Erratic monsoons, temperature extremes, and shifting seasons disrupt traditional forecasting
- **Precision agriculture:** Data-driven decisions can reduce waste and optimise inputs

Why Machine Learning?

Non-linear Patterns

Captures complex agro-climatic interactions invisible to statistical models

Multi-feature Fusion

Integrates climate, soil, and management data simultaneously

Scalability

Generalises across crops, states, and seasons without re-parameterisation





Research Objectives

1

Compare 6 ML Algorithms

Benchmark Ridge Regression, Decision Tree, Random Forest, SVR, XGBoost, and Gradient Boosting under identical conditions

2

Predict Crop Yield Across Indian States

Cover 15 states, 6 major crops, and 26 years of historical agro-climatic data

3

Evaluate Model Accuracy

Use MAE, RMSE, R^2 , MAPE, and CV- R^2 for rigorous, multi-metric evaluation

4

Analyse Feature Importance

Identify the most predictive agro-climatic, soil, and management features

5

Build Reproducible Benchmark

Release a standardised pipeline for future Indian agricultural ML research



Thank You